CIE 3 MINI PROJECT- ECOMMERCE MANAGEMENT SYSTEM

Submitted by

NAME: Ninada S A, Sanajna Pal, Sahasra Marreboyin

USN: 1RUA25BCA0075, 1RUA25BCA0096, 1RUA25BCA0093

as part of the ***Continuous Internal Evaluation*** *for the course*

CS2201 DATABASE MANAGEMENT SYSTEM

Offered in

Semester III

of

BCA (Hons)



School of Computer Science and Engineering
RV University
RV Vidyaniketan,8th Mile, Mysore Road,
Bengaluru, Karnataka, India - 560059

October 2026

SQL CODE:

```
CREATE DATABASE ecommerce_db;

USE ecommerce_db;



CREATE TABLE  customers (

    customer_id INT PRIMARY KEY,

    name VARCHAR(100),

    email VARCHAR(100),

    phone VARCHAR(15)

);

CREATE TABLE  products (

    product_id INT PRIMARY KEY,

    product_name VARCHAR(100),
```

```
    price DECIMAL(10, 2),
    stock INT
);

CREATE TABLE  orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    total_amount DECIMAL(10, 2)
);
INSERT INTO customers (customer_id, name, email, phone) VALUES
(1, 'Aarav Sharma', 'aarav@gmail.com', '9876543210'),
(2, 'Priya Patel', 'priya@gmail.com', '9123456789');

INSERT INTO products (product_id, product_name, price, stock) VALUES
(101, 'Wireless Mouse', 799.00, 50),
(102, 'Mechanical Keyboard', 2499.00, 30),
(103, 'USB-C Hub', 1299.00, 40);
```

INSERT INTO orders (order_id, customer_id, order_date, total_amount) VALUES

(1001, 1, '2026-10-31', 799.00),

(1002, 2, '2026-10-31', 2499.00);

```sql
CREATE DATABASE ecommerce_db;
USE ecommerce_db;

CREATE TABLE  customers (
    customer_id INT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100),
    phone VARCHAR(15)
);
CREATE TABLE  products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    price DECIMAL(10, 2),
    stock INT
);

CREATE TABLE  orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    total_amount DECIMAL(10, 2)
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INSERT INTO orders (order_id, customer_id, order_date, total_amount) VALUES
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(1002, 2, '2026-10-31', 2499.00);
```

JAVA CODE:

```java
 * @author Ninada S A
 */
import java.sql.*;
import java.util.Scanner;

/**

 */
public class Ecommerece {
    public static void main(String[] args) {
        String url = "jdbc:mysql://localhost:3306/ecommerce_db";
        String user = "root";
        String password = "urpwd";

        try {

            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();
            Scanner scanner = new Scanner(System.in);
            int choice;
```

```
do {
    System.out.println("\n--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---");
    System.out.println("1. View All Products");
    System.out.println("2. Insert Product (Create)");
    System.out.println("3. View All Customers");
    System.out.println("4. Insert Customer (Create)");
    System.out.println("5. View All Orders");
    System.out.println("6. Insert Order (Create)");
    System.out.println("7. Update Product Price (Update)");
    System.out.println("8. Delete Product (Delete)");
    System.out.println("9. Exit");
    System.out.print("Enter your choice: ");
    choice = scanner.nextInt();

    switch(choice) {
        case 1:
            ResultSet rsProd = stmt.executeQuery("SELECT * FROM products");
            System.out.println("\n--- Products List ---");
            while(rsProd.next()) {
                System.out.println("ID: " + rsProd.getInt("product_id") +
                                   " | Name: " + rsProd.getString("product_name") +
                                   " | Price: " + rsProd.getDouble("price") +
                                   " | Stock: " + rsProd.getInt("stock"));
            }
            rsProd.close();
            break;
```

```
        case 2:
            System.out.print("Enter Product ID: ");
            int pid = scanner.nextInt();
            scanner.nextLine();
            System.out.print("Enter Product Name: ");
            String pname = scanner.nextLine();
            System.out.print("Enter Price: ");
            double price = scanner.nextDouble();
            System.out.print("Enter Stock: ");
            int stock = scanner.nextInt();

            stmt.executeUpdate("INSERT INTO products VALUES (" + pid + ", '" + pname + "', " + price + "
            System.out.println("Product inserted successfully.");
            break;

        case 3:
            ResultSet rsCust = stmt.executeQuery("SELECT * FROM customers");
            System.out.println("\n--- Customers List ---");
            while(rsCust.next()) {
                System.out.println("ID: " + rsCust.getInt("customer_id") +
                                   " | Name: " + rsCust.getString("name") +
                                   " | Email: " + rsCust.getString("email") +
                                   " | Phone: " + rsCust.getString("phone"));
            }
            rsCust.close();
            break;
```

```
            case 4:
                System.out.print("Enter Customer ID: ");
                int cid = scanner.nextInt();
                scanner.nextLine();
                System.out.print("Enter Name: ");
                String cname = scanner.nextLine();
                System.out.print("Enter Email: ");
                String email = scanner.nextLine();
                System.out.print("Enter Phone: ");
                String phone = scanner.nextLine();

                stmt.executeUpdate("INSERT INTO customers VALUES (" + cid + ", '" + cname + "', '" + email
                System.out.println("Customer inserted successfully.");
                break;

            case 5:
                ResultSet rsOrd = stmt.executeQuery("SELECT * FROM orders");
                System.out.println("\n--- Orders List ---");
                while(rsOrd.next()) {
                    System.out.println("Order ID: " + rsOrd.getInt("order_id") +
                                       " | Customer ID: " + rsOrd.getInt("customer_id") +
                                       " | Date: " + rsOrd.getDate("order_date") +
                                       " | Total: " + rsOrd.getDouble("total_amount"));
                }
                rsOrd.close();
                break;
```

```
            case 6:
                System.out.print("Enter Order ID: ");
                int oid = scanner.nextInt();
                System.out.print("Enter Customer ID: ");
                int ocid = scanner.nextInt();
                scanner.nextLine();
                System.out.print("Enter Order Date (YYYY-MM-DD): ");
                String odate = scanner.nextLine();
                System.out.print("Enter Total Amount: ");
                double total = scanner.nextDouble();

                stmt.executeUpdate("INSERT INTO orders VALUES (" + oid + ", " + ocid + ", '" + odate + "', "
                System.out.println("Order inserted successfully.");
                break;

            case 7:
                System.out.print("Enter Product ID to update: ");
                int upid = scanner.nextInt();
                System.out.print("Enter New Price: ");
                double nprice = scanner.nextDouble();

                int rowsUpdated = stmt.executeUpdate("UPDATE products SET price = " + nprice + " WHERE produ
                if(rowsUpdated > 0) System.out.println("Product price updated successfully.");
                else System.out.println("Product ID not found.");
                break;
```

```
                    case 8:
                        System.out.print("Enter Product ID to delete: ");
                        int dpid = scanner.nextInt();

                        int rowsDeleted = stmt.executeUpdate("DELETE FROM products WHERE product_id = " + dpid);
                        if(rowsDeleted > 0) System.out.println("Product deleted successfully.");
                        else System.out.println("Product ID not found.");
                        break;

                    case 9:
                        System.out.println("Exiting application.");
                        break;

                    default:
                        System.out.println("Invalid choice. Try again.");
                }
            } while(choice != 9);


            scanner.close();
            stmt.close();
            con.close();
        }
        catch(Exception e) {
            e.printStackTrace();
        }
    }
}
```

OUTPUT:

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 1

--- Products List ---
ID: 101 | Name: Wireless Mouse | Price: 799.0 | Stock: 50
ID: 102 | Name: Mechanical Keyboard | Price: 2499.0 | Stock: 30
ID: 103 | Name: USB-C Hub | Price: 1299.0 | Stock: 40
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 2
Enter Product ID: 106
Enter Product Name: lamp
Enter Price: 670
Enter Stock: 87
Product inserted successfully.
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 3

--- Customers List ---
ID: 1 | Name: Aarav Sharma | Email: aarav@gmail.com | Phone: 9876543210
ID: 2 | Name: Priya Patel | Email: priya@gmail.com | Phone: 9123456789
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 4
Enter Customer ID: 5
Enter Name: priya
Enter Email: priya2123
Enter Phone: 890756386
Customer inserted successfully.
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 5

--- Orders List ---
Order ID: 1001 | Customer ID: 1 | Date: 2026-10-31 | Total: 799.0
Order ID: 1002 | Customer ID: 2 | Date: 2026-10-31 | Total: 2499.0
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 6
Enter Order ID: 106
Enter Customer ID: 5
Enter Order Date (YYYY-MM-DD): 7-6-26
Enter Total Amount: 67000
Order inserted successfully.
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 7
Enter Product ID to update: 106
Enter New Price: 677
Product price updated successfully.
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 8
Enter Product ID to delete: 106
Product deleted successfully.
```

```
--- E-COMMERCE MANAGEMENT SYSTEM (ALL TABLES) ---
1. View All Products
2. Insert Product (Create)
3. View All Customers
4. Insert Customer (Create)
5. View All Orders
6. Insert Order (Create)
7. Update Product Price (Update)
8. Delete Product (Delete)
9. Exit
Enter your choice: 9
Exiting application.
BUILD SUCCESSFUL (total time: 1 minute 59 seconds)
```